Minia University

Faculty of Engineering

Computers and Systems Engineering Department

IOT Project Proposal

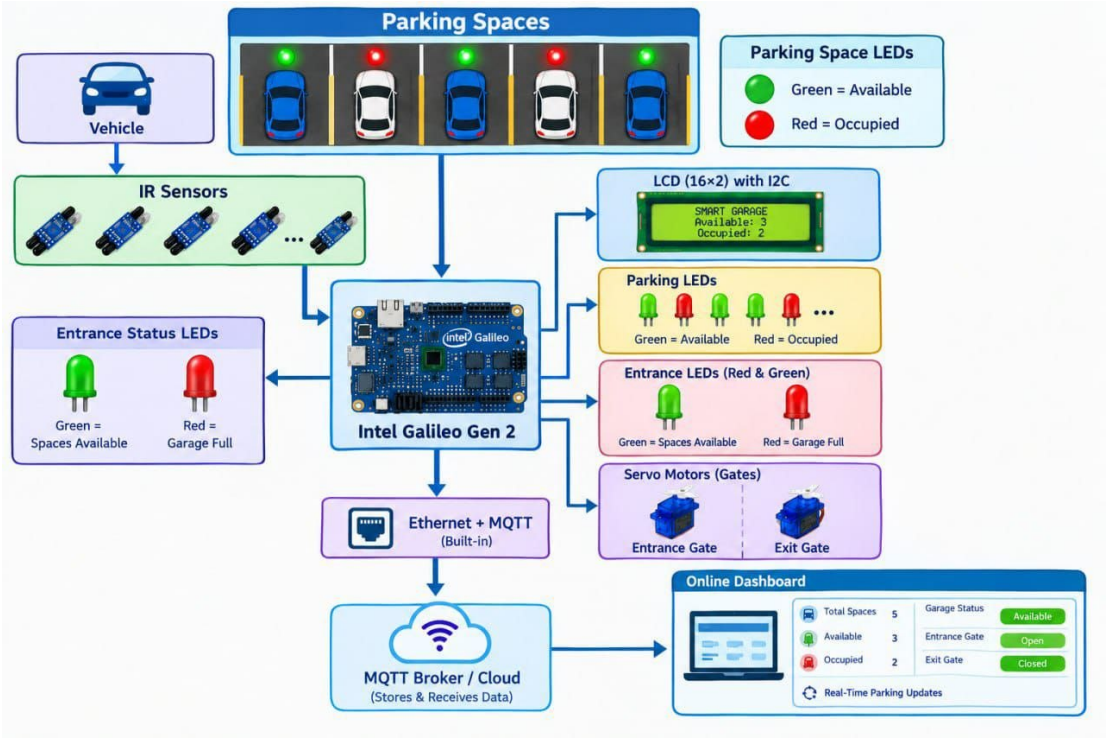
Smart Automated Garage System

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1. Problem / Introduction

Parking garages often face problems such as difficulty finding available parking spaces, long waiting times at entrances, and inefficient parking management. The proposed Smart Automated Garage System uses IoT technologies to monitor parking occupancy, automate gate operation, and provide real-time parking information through an online dashboard, improving efficiency and user convenience.

2. Objectives

- Detect occupied and available parking spaces using IR sensors.
- Automatically control entrance and exit gates.
- Display parking availability on an LCD screen.
- Indicate parking space status using LEDs.
- Indicate garage availability using red and green entrance LEDs.
- Transmit parking information to an online dashboard in real time.
- Enable remote monitoring through Internet connectivity.
- Improve garage management and reduce vehicle waiting time.

3. Proposed IoT Architecture

Physical Layer

Includes vehicles, parking spaces, and garage gates.

Sensors & Actuators Layer

- IR Sensors: Detect vehicle presence.
- SG90 Servo Motors: Control entrance and exit gates.
- Parking LEDs: Indicate whether a parking space is occupied or available.
- Entrance LEDs: Green indicates available spaces; red indicates a full garage.
- LCD: Displays parking information.

Device / Edge Layer

The Intel Galileo Gen 2 collects sensor data, processes it, calculates available parking spaces, and controls the LEDs, LCD, and servo motors.

Connectivity Layer

The Intel Galileo transmits data through its built-in Ethernet interface using the MQTT protocol.

Platform / Backend Layer

An MQTT broker/cloud platform receives and stores parking information.

Application Layer

Users monitor the garage through an online dashboard displaying parking availability and gate status in real time.

4. Main Components

- Intel Galileo Gen 2
- IR Sensors
- SG90 Servo Motors (2)
- 16×2 LCD with I2C Module
- Parking Space LEDs
- Entrance Status LEDs (Red & Green)
- Resistors
- Breadboard
- Jumper Wires
- Ethernet Interface (Built-in)
- MQTT Protocol

5. Data & Control Flow

Data Flow

Vehicle → IR Sensor → Intel Galileo → Ethernet → MQTT Broker → Dashboard

IR sensors detect vehicle presence in parking spaces. The Intel Galileo processes the sensor readings and updates the number of available spaces. Parking data is then transmitted through Ethernet using MQTT, and the dashboard displays the updated garage status.

Control Flow

System Decision → Intel Galileo → LEDs & Servo Motors

The Intel Galileo calculates the number of available parking spaces.

If spaces are available:

- Green entrance LED turns ON.
- Red entrance LED turns OFF.

Entrance gate can open.

If the garage is full:

- Red entrance LED turns ON.
- Green entrance LED turns OFF.

Entrance gate remains closed.

Servo motors automatically control the opening and closing of the entrance and exit gates.

6. Expected Output / User Interface

LCD Display

SMART GARAGE

Available: 3

Occupied: 2
Parking Space LEDs

- Green LED → Available Space
- Red LED → Occupied Space
  Entrance Status LEDs
- Green LED ON → Parking spaces available.
- Red LED ON → Garage full.

Online Dashboard
The dashboard displays:

- Total Parking Spaces
- Available Spaces
- Occupied Spaces
- Garage Status (Available / Full)
- Entrance Gate Status
- Exit Gate Status
- Real-Time Parking Updates

7. References

YouTube. Arduino: Smart Parking System Project Tutorial. Available at: https://youtu.be/hl9KllEYszA?si=3y0Yh_IwABQZRaPM

OpenAI. ChatGPT. Used as an AI-assisted tool for brainstorming, organizing, and reviewing the project proposal.